

DATABASE SYSTEMS

Course Code: CS-151

Credit Hours: 4(3+1)

Pre-requisite: Nil

Course Objectives

The key objectives of this course include:

- List and explain the terminology of the relational database model
- Analyze business information requirements using entity relationship diagrams (ERD)
- Analyze a data model (ERD) into a logical database design with normalized relations
- Design, develop, test and prove the functionality of a database
- Construct, write and execute SQL queries
- Explain the necessary steps to develop an effective physical database design

Course Learning Outcomes

After successful completion of this course, the students should be able to: GA BT Level

1. Explain fundamental database concepts	2	C2
2. Identify functional dependencies and Produce SQL queries for database definition and manipulation in any DBMS	3	C3
3. Discover conceptual, logical and physical database schemas using different data models	4	C4

Course Contents

Basic database concepts, Database approach vs. file based system, database architecture, three level schema architecture, data independence, relational data model, attributes schemas, tuples, domains, relation instances, keys of relations, integrity constraints, relational algebra, selection, projection, Cartesian product, types of joins, normalization, functional dependencies, normal forms, entity relationship model, entity sets, attributes, relationship, entity-relationship diagrams, Structured Query Language (SQL), Joins and subqueries in SQL, Grouping and aggregation in SQL, concurrency control, database backup and recovery, indexes, NoSQL systems.

Mapping of CLOs to GA's

GA's/CLOs	CLO1	CLO2	CLO3	CLO4
GA1 (Academic Education)				
GA2 (Knowledge for Solving Computing Problems)	✓			
GA3 (Problem Analysis)			✓	
GA4 (Design/Development of Solutions)		✓		
GA5 (Modern Tool Usage)				
GA6 (Individual and Teamwork)				
GA7 (Communication)				
GA8 (Computing Professionalism and Society)				
GA9 (Ethics)				
GA10 (Life-long Learning)				

Resources

Text Book:

Database Systems: A Practical Approach to Design, Implementation, and Management by Thomas Connolly and Carolyn Begg 6th Edition, 2021

Database Management Systems by Raghu Ramakrishnan, Johannes Gehrke, 3rd Edition 2002

Database System Concepts by Avi Silberschatz, Henry F. Korth and S. Sudarshan, 7th Edition, 2019

Database Systems: The Complete Book by Hector Garcia-Molina, Jeffrey D. Ullman, Jennifer

Wido, 2nd Edition, 2008

Weekly Course Plan

Week	Topics
Week 1	Introduction to Databases • Traditional File-Based Systems • File-Based Approach • Limitations of the File-Based Approach • Database Approach • The Database • The Database Management System (DBMS) • (Database) Application Programs • Components of the DBMS Environment • Database Design: The Paradigm Shift • Roles in the Database Environment • Data and Database Administrators • Database Designers • Application Developers • End-Users • History of Database Management Systems • Advantages and Disadvantages of DBMSs
Week 2	Database Environment • The Three-Level ANSI-SPARC Architecture • External Level • Conceptual Level • Internal Level • Schemas, Mappings, and Instances • Data Independence
Week 3	Database Languages • The Data Definition Language (DDL) • The Data Manipulation Language (DML)

	• Fourth-Generation Languages (4GLs) • Data Models and Conceptual Modeling • Object-Based Data Models • Record-Based Data Models • Physical Data Models • Conceptual Modeling • Functions of a DBMS
Week 4	Database Architectures and the Web • Multi-user DBMS Architectures • Teleprocessing • File-Server Architecture • Traditional Two-Tier Client–Server Architecture • Three-Tier Client–Server Architecture • N-Tier Architectures
Week 5	The Relational Model 149 • Brief History of the Relational Model • Terminology • Relational Data Structure • Mathematical Relations • Database Relations • Properties of Relations • Relational Keys • Representing Relational Database Schemas • Integrity Constraints • Nulls • Entity Integrity • Referential Integrity • General Constraints • Views • Terminology • Purpose of Views • Updating Views
Week 6	Database System Development Lifecycle • The Information Systems Lifecycle • The Database System Development Lifecycle • Database Planning • System Definition • User Views • Requirements Collection and Analysis • Centralized Approach • View Integration Approach • Database Design

	• Approaches to Database Design • Data Modeling • Phases of Database Design
Week 7	Entity–Relationship Modeling • Entity Types • Relationship Types • Degree of Relationship Type • Recursive Relationship • Attributes • Simple and Composite Attributes • Single-valued and Multi-valued Attributes • Derived Attributes • Keys • Strong and Weak Entity Types • Attributes on Relationships
Week 8	• Structural Constraints • One-to-One (1:1) Relationships • One-to-Many (1:*) Relationships • Many-to-Many (*:*) Relationships • Cardinality and Participation Constraints • Problems with ER Models
MID SEMESTER EXAMS	
Week 9& 10	Enhanced Entity–Relationship Modeling • Specialization/Generalization • Superclasses and Subclasses • Superclass/Subclass Relationships • Attribute Inheritance • Specialization Process • Generalization Process • Constraints on Specialization/Generalization • Aggregation • Composition
Week 11	Normalization • The Purpose of Normalization • How Normalization Supports Database Design • Data Redundancy and Update Anomalies • Insertion Anomalies

	• Deletion Anomalies • Modification Anomalies
Week 12	Functional Dependencies • Characteristics of Functional Dependencies • Identifying Functional Dependencies • Identifying the Primary Key for a Relation • Using Functional Dependencies • The Process of Normalization • First Normal Form (1NF) • Second Normal Form (2NF) • Third Normal Form (3NF) • General Definitions of 2NF and 3NF
Week 13	SQL: Data Manipulation • Introduction to SQL • Objectives of SQL • History of SQL • Importance of SQL • Terminology • Writing SQL Commands
Week 14	• Data Manipulation • Simple Queries • Sorting Results (ORDER BY Clause) • Using the SQL Aggregate Functions • Grouping Results (GROUP BY Clause)
Week 15	Sub queries • ANY and ALL • Multi-table Queries • EXISTS and NOT EXISTS • Combining Result Tables (UNION, INTERSECT, EXCEPT) • Database Updates
Week 16	Ethical and Security issues in Database
END SEMESTER EXAMS	

DATABASE SYSTEMS LAB

Course Code:CS-151L

LAB Objectives

The objective of this lab course is to understand the practical applicability of database and its management system concepts. DBMS lab aims at practicing and achieving this aim by using various software's such as MS Access, SQL Server. All these require thorough practice of various SQL queries. Working on existing database systems, designing of database, creating relational database, and analysis of table design. The lab course also provides practical knowledge to understand advanced database concepts such as Structured Query Language. Joins and sub-queries in SQL, Grouping and aggregation in SQL, concurrency control, database backup and recovery, indexes, NoSQL systems.

Lab Learning Outcomes

After successful completion of this course, the students should be able to: GA BT Level

1. Practice key components of relational database systems.	2	P3
2. Produce database and SQL (structured query language) based upon a data model	5	P4
3. Express your solutions from problems conducted during your project report.	6	A3

Lab Contents

User interface, data independence, user view, three data models (relational, hierarchical, network, object oriented), conceptual, logical and physical database design and evaluation, normalization, query languages, query optimization, security, integrity and concurrency protocols, introduction to SQL and its application to RDBMS. Database design, model building, data table, forms & reports. Database administration.

Mapping of LLOs to GA's

GA's/LLOs	LLO1	LLO2	LLO3
GA1 (Academic Education)			
GA2 (Knowledge for Solving Computing Problems)	✓		
GA3 (Problem Analysis)			
GA4 (Design/Development of Solutions)			
GA5 (Modern Tool Usage)		✓	
GA6 (Individual and Teamwork)			✓
GA7 (Communication)			
GA8 (Computing Professionalism and Society)			
GA9 (Ethics)			
GA10 (Life-long Learning)			

Resources

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2019 Database Systems: The Complete Book by Hector Garcia-Molina, Jeffrey D. Ullman, Jennifer Wido, 2nd on 2008


 Prof. Dr. Asmita Hanit
 Director Academics, Johnson Studies & Research
 University Of Wollongong

Weekly List of Experiments			
Week no.	Topics	LLO, GA	
1	Guidance to MS Access	LLO1, GA2	
2	Practice validation Rules on MS Access	LLO1, GA2	
3	Practice creating forms and reports on MS Access	LLO1, GA2	
4	Execute SQL in Access	LLO2, GA5	
5	Practice creating Entity-Relationship Diagrams	LLO1, GA2	
6	Practice creating Enhanced Entity-Relationship Diagrams	LLO1, GA2	
7	Guidance of MS SQL Server 2019	LLO1, GA2	
8	Lab Sessional	LLO1, GA2	
9	Execute DML Commands	LLO2, GA5	
10	Execute Where Clause and Data Sorting	LLO2, GA5	
11	Execute Data Restriction and Constraints	LLO2, GA5	
12	Open ended lab	LLO3, GA6	
13	Execute Between Operator and Setting domain, entity and referential constraints	LLO2, GA5	
14	Execute Aggregating and Scalar Functions	LLO2, GA5	
15	Execute Joins in SQL	LLO2, GA5	
16	Lab Sessional	LLO2, GA5	